

EDUCATION	Department of Earth Sciences, Dartmouth College	Hanover, NH
	<i>Ph.D. Research Scholar</i>	2020 - present
	• Advisor: Prof. C. Brenhin Keller • Thesis: Insights Into the Volcanic-Plutonic Connection at Convergent Plate Margins From Global Petrogeochemistry and Mineral Thermobarometry • Research area: Geochemistry, Petrology	
	School of Earth Sciences, China University of Geosciences	Wuhan, China
	<i>Ph.D. in Geosciences</i>	2017 - 2022
	• Advisor: Prof. Changqian Ma • Dissertation successfully defended Oct 2022 Degree awarded Dec 2024	
	School of Earth Sciences, China University of Geosciences	Wuhan, China
	<i>M.S. in Geology</i>	2015 - 2017
	School of Earth Sciences, China University of Geosciences	Wuhan, China
	<i>B.S. in Geology (Honors)</i>	2011 - 2015
PROJECTS	Probing the Growth Processes of Triassic Granitic Plutons in the East Kunlun Orogenic Belt and Their Coupling/Decoupling Mechanisms with Volcanic Activity	
	<i>National Natural Science Foundation of China</i>	2020 - 2024
	• Conducted thermobarometric and thermodynamic modeling analyses. • Leveraged high-performance computing for data processing.	
	Investigation of Crystal Mush Activation Processes in Transcrustal Magmatic Systems Based on the Diversity of Felsic Rocks	
	<i>National Natural Science Foundation of China</i>	2019 - 2022
	• Conducted field investigations and igneous petrology studies. • Integrated geochemistry and geochronology to study magmatic processes.	
	Geological Mapping of Mixed Granites in the Jialuhe Area, Eastern Segment of the East Kunlun	
	<i>China Geological Survey</i>	2015 - 2017
	• Created detailed geologic maps for tectonic analysis. • Investigated petrogenesis and tectonic evolution.	
SKILLS	Programming: Python, Julia, R, MATLAB, C.	
	Technical Expertise: Data Science, Thermodynamic Modeling, Causal Inference, High-Performance Computing, Machine Learning.	
	Languages: Chinese (native), English (fluent)	
AWARDS AND SCHOLARSHIPS	• Joint Education Scholarship , China University of Geosciences	2022
	• National Scholarship for Studying Abroad , China Scholarship Council	2021 - 2022
	• Joint Training Scholarship , China University of Geosciences	2019
	• Graduate Study Scholarship , China University of Geosciences	2017 - 2021
	• National Fundamental Sciences Outstanding Talent Scholarship , China University of Geosciences	2011 - 2015

CONFERENCES	Insights Into Magma Storage and Reequilibration From Hornblende Thermobarometry <i>Goldschmidt, oral</i>	Hawaii, 2022
	Terminal Porosity and the Geochemical Detectability of Plutonic Residues of Melt Extraction <i>Goldschmidt, oral</i>	Online, 2020
TEACHING	Petrology <i>Teaching assistant, Prof. Changqian Ma</i>	Fall 2017
	Advanced Petrology <i>Teaching assistant, Prof. Jaroslav Dostal</i>	Fall 2016
PUBLICATIONS	1. Ke Gao, C. Brenhin Keller, Changqian Ma. Amphibole Compositions Record Cold Post-emplacement Reequilibration in Plutons. <i>Journal of Geophysical Research: Solid Earth</i>, 2025. 2. Guizhi Huang, Changqian Ma, Bin Liu, Zhenhua Xue, Huai Xiong, Kaipei Lu, Quancheng Lian, Ke Gao, Mengyu Zhang, Siyu Yang. Apatite and Zircon as key indicators for effusive-explosive transition in the Leningcou Volcano: Middle Triassic Volcanic activity in the East Kunlun Orogenic Belt. <i>Chemical Geology</i>, 2025 3. Ke Gao, C. Brenhin Keller, Changqian Ma. Can Geochemistry Distinguish Extracted Melt from Cumulate with Trapped Melt? <i>Journal of Earth Science</i>, 2024. 4. Changqian Ma, Bin Liu, Zhenhua Xue, Bowen Zou, Guizhi Huang, Quancheng Lian, Ke Gao, Yang Sun, Lianxun Wang. Insights Into the Volcanic Eruption Process and Magmatic Plumbing Systems from Felsic Intrusions. <i>Acta Petrologica Sinica</i>, 2024. 5. Changqian Ma, Bin Liu, Bowen Zou, Ke Gao. How to Gain an Insight into the Volcano's Eruption Mechanism from the Study of Magmatic Intrusions? <i>Earth Science</i>, 2022. 6. Changqian Ma, Bowen Zou, Ke Gao, Xia Wen. Crystal Mush Storage, Incremental Pluton Assembly and Granitic Petrogenesis. <i>Earth Science</i>, 2020. 7. Bin Liu, Changqian Ma, Pan Guo, Yang Sun, Ke Gao, Yuheng Guo. Evaluation of Late Permian Mafic Magmatism in the Central Tibetan Plateau as a Response to Plume-Subduction Interaction. <i>Lithos</i>, 2016. 8. Changqian Ma, Fuhao Xiong, Shuo Yin, Lianxun Wang, Ke Gao. Intensity and Cyclicity of Orogenic Magmatism: An Example From a Paleo-Tethyan Granitoid Batholith, Eastern Kunlun, Northern Qinghai-Tibetan Plateau. <i>Acta Petrologica Sinica</i>, 2015.	